THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 3

Chapters 9–11: **Life Processes in Plants, Light: Shadows and Reflections, Earth, Moon and the Sun** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A plant wilts when it loses more water than it absorbs. This is because: 1

(a) it makes more food **(b)** cells lose water and become limp
(c) it gets more light **(d)** it grows

2. The shadow of a pole in the afternoon falls towards the: 1

(a) east **(b)** west **(c)** north only **(d)** south only

3. The first quarter moon is seen as: 1

(a) a dark disc
(b) thin crescent
(c) full disc
(d) half disc, right half lit (northern hemisphere)

4. In lichen, the alga: 1

(a) fixes nitrogen **(b)** makes food by photosynthesis
(c) gives shelter **(d)** attacks the fungus

5. A pinhole camera cannot form an image if: 1

(a) the screen is translucent **(b)** the object is bright
(c) the hole is small **(d)** the hole is covered

6. During a total lunar eclipse the Moon appears: 1

(a) dark reddish **(b)** green **(c)** blue **(d)** bright white

7. Plants kept in sunlight release bubbles of oxygen in water. A hydrilla plant in a funnel shows: 1

(a) no bubbles **(b)** bubbles only in dark
(c) constant bubbles **(d)** more bubbles in sunlight

8. A boy 1.2 m tall stands before a plane mirror. His image is: 1

(a) 1.2 m tall **(b)** 0 m **(c)** 0.6 m tall **(d)** 2.4 m tall

9. The length of day on the equator throughout the year is about: 1

(a) 18 hours **(b)** 24 hours **(c)** 6 hours **(d)** 12 hours

10. Plants that grow in deserts have fewer stomata or spines to: 1

(a) increase CO2 **(b)** reduce water loss
(c) increase light **(d)** increase food

11. A lamp is moved away from an object. The shadow size: 1

(a) stays same **(b)** increases **(c)** vanishes **(d)** decreases

12. Half the Moon's surface is always lit by the Sun because: 1

(a) the Earth shines on it **(b)** the Sun always shines on half of it

(c) stars light it **(d)** it makes light

13. In lichen, the fungus: 1

(a) eats insects **(b)** makes food

(c) makes oxygen **(d)** gives shelter and absorbs water

14. Which of these is formed on the screen of a pinhole camera? 1

(a) virtual image **(b)** real image

(c) shadow of the hole **(d)** a reflection

15. Which eclipse can be watched safely with the naked eye? 1

(a) partial solar only **(b)** lunar

(c) both of them **(d)** solar

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. A girl is 2 m from a plane mirror. If she moves 1 m closer, what is the distance between her and her image? 2

17. Why do places near the equator have almost no seasonal change? 2

18. Why does Cuscuta not need chlorophyll? 2

19. How would you find the height of a tree using its shadow and a stick? 2

20. Explain the term 'lunar month'. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Explain how seasons affect agriculture in India. 3

22. Explain why a field of pulses needs less fertiliser. 3

23. Describe the use of a periscope in submarines. 3

24. Why is the sky dark during a total solar eclipse? 3

OR

Explain how the pitcher plant is adapted for trapping insects. 3

25. Describe how to make a pinhole camera and observe the image of a candle. 3

OR

Compare the day length in Delhi in June and December and give the reason. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Explain with a labelled description how the phases of the Moon are caused and name eight phases in order. 5

OR

Explain the relationship of lichen and Rhizobium as examples of symbiosis. 5

MASTERMIND

27. A girl stands 2 m in front of a mirror and steps back by 1 m. Find the new distance between her and her image. 5

OR

Discuss the importance of the Moon for Earth: night light, calendars, festivals and study of space. 5

END OF QUESTION PAPER **Total: 50 Marks**

MASTERMIND